

FEDERAL BOARD OF INTERMEDIATE & SECONDARY EDUCATION

Annual Examination 2026 — ANSWER KEY (Version 1)

Chemistry — Class 9 (Science Group)

Syllabus: Chapter 1 — Nature of Chemistry | Chapter 2 — Matter | Chapter 3 — Atomic Structure | Chapter 4 — Periodic Table & Periodicity | Chapter 5 — Chemical Bonding | Chapter 6 — Stoichiometry | Chapter 7 — Electrochemistry | Chapter 8 — Energetics | Chapter 9 — Chemical Equilibrium

Total Marks: 46

Time: 2 Hours

Paper: Exam-21 (Version 1)

SECTION A — Multiple Choice Questions ($30 \times 1 = 30$ Marks)

Q1. Which branch of chemistry mainly studies the energy changes that accompany chemical reactions? → **A – Physical chemistry** 1 mark

Physical chemistry is the branch that deals with **energy changes, structure and rates** of chemical processes, distinguishing it from analytical (composition/measurement), industrial (large-scale manufacturing) and nuclear (radioactive) chemistry.

SLO: Ch1 p.1 — explain that chemistry has many sub-fields. Source: Ch1, branches-of-chemistry chapter opener.

Q2. Which statement best distinguishes technology from science? → **B – Technology applies scientific knowledge to solve practical, everyday problems** 1 mark

Science seeks to **understand** natural phenomena, while technology **applies** that understanding to build practical tools and solutions. Option A is false since technology also relies on tested science; C confuses science's role; D wrongly equates the two.

SLO: Ch1 p.1 — differentiate between 'science', 'technology' and 'engineering'.

Q3. A mixture whose particles are too small to settle but large enough to scatter light (Tyndall effect) is a: → **C – Colloid** 1 mark

A **colloid** has particle sizes between a true solution and a suspension; these particles scatter light (Tyndall effect) but do not settle out on standing, unlike a suspension.

SLO: Ch2 p.1 — identify solutions, colloids and suspensions as mixtures.

Q4. Which is an example of a solid solute dissolved in a liquid solvent forming a true solution? → **D – Sugar dissolved in water** 1 mark

Sugar (solid) dissolves completely and uniformly in water (liquid) to give a **true solution**. Smoke-in-air is a solid-in-gas colloid, milk is a liquid-in-liquid colloid (emulsion), and muddy water is a suspension.

SLO: Ch2 p.1 — identify solutions, colloids and suspensions and give an example of each.

Q5. Diamond, graphite and fullerenes are different forms of carbon. These are called: → **B – Allotropes** 1 mark

Allotropes are different structural forms of the SAME element existing in the same physical state. Diamond, graphite and fullerenes are all solid carbon but with different bonding arrangements.

SLO: Ch2 p.1 — explain the allotropic forms of solids (examples include diamond, graphite and fullerenes).

Q6. An atom as a whole is electrically neutral because: → **A – the number of protons equals the number of electrons** 1 mark

Protons carry a +1 charge and electrons a –1 charge; in a neutral atom these counts are **equal**, so the positive and negative charges exactly balance, giving no net charge. Neutrons carry no charge and play no role in this balance.

SLO: Ch3 p.1 — state the relative charge of subatomic particles; interpret the relationship between a subatomic particle, its mass and charge. Source: Ch3 p.33, heading "Charge Neutrality" — "Atoms are electrically neutral because the number of protons... is equal to the number of electrons... The charges balance each other so there is no net charge on the atoms."

Q7. An atom of zinc (atomic number 30, mass number 65). Number of neutrons? → **D – 35** 1 mark

Number of neutrons = mass number – atomic number = $65 - 30 = 35$.

SLO: Ch3 p.2 — determine the number of protons and neutrons of different isotopes. Source: Ch3 p.33, "No. of neutrons = mass number – atomic number".

Q8. Which statement correctly describes isotopes of the same element? → **C – Same number of protons but a different number of neutrons** **1 mark**

Isotopes of an element always have the same atomic number (same protons, keeping them the same element) but differ in mass number because they have different numbers of neutrons.

SLO: Ch3 p.2 — define isotopes as different atoms of the same element that have the same number of protons but different neutrons.

Q9. The Quantum Mechanical Model describes the electron's position in terms of: → **A – probability regions called orbitals** **1 mark**

The modern quantum mechanical model treats electrons as wave-particles whose exact path cannot be known (Heisenberg's uncertainty principle); instead, it defines **orbitals** — regions of space where an electron is most likely to be found.

SLO: Ch3 p.1 — state the electrons are quantum particles with probabilistic paths whose exact location cannot be mapped. Source: Ch3 p.31, "Quantum Mechanical Model".

Q10. Transition elements show variable oxidation states mainly because: → **B – both their s and d sub-shell electrons can take part in bonding** **1 mark**

Because the energies of the **(n-1)d and ns sub-shells** are close, transition elements can use different numbers of electrons from both sub-shells in bonding, giving rise to multiple oxidation states, e.g. iron shows +2 and +3, copper shows +1 and +2.

SLO: Ch4 p.2 — describe the transition elements as metals that have variable oxidation numbers. Source: Ch4 p.64, §4.5 "Transition Elements", point 3 "Variable Oxidation States" — "This is because of d-sub shell can also participate in bonding along with s-sub shell."

Q11. The LENGTH of a period is determined mainly by: → **D – the type of sub-shell (s, p, or d) being filled across that period** **1 mark**

Each period ends once the sub-shells available at that energy level are completely filled; since s, p and d sub-shells hold different numbers of electrons, the **type of sub-shell filling** across a period fixes how many elements that period contains.

SLO: Ch4 p.1 — identify the group or period or block of an element using its electronic configuration. Source: Ch4 p.68, Key Points — "The length of a period in the periodic table depends on the type of sub-shell that fills."

Q12. Group 1 and Group 2 elements are together called the: → **C – s-block elements** **1 mark**

Elements of Group 1 and Group 2 have their **outermost (valence) electrons filling the s sub-shell**, which is why they are together classified as s-block elements.

SLO: Ch4 p.1 — identify the block of an element using its electronic configuration. Source: Ch4 p.68, Key Points — "Group 1 and 2 elements are called s-block elements, since s sub-shell fills in these elements."

Q13. Electron affinity is best defined as: → **B – the energy released when an isolated gaseous atom gains an electron to form a uni-negative ion** **1 mark**

Electron affinity specifically concerns an atom **gaining** an electron and the energy **released** in the process, unlike ionization energy, which concerns removing an electron.

SLO: Ch4 p.1 — identify trends in groups and periods including electron affinity. Source: Ch4 p.68, Key Points — "Electron affinity is the amount of energy released when an electron adds up in the valence shell of an isolated atom to form a uni-negative gaseous ion."

Q14. Which transition metal is used as the catalyst in the Haber process? → **A – Iron** **1 mark**

Iron is used as the industrial catalyst for the Haber process (synthesis of ammonia from nitrogen and hydrogen). Platinum/palladium are used in catalytic converters, and nickel in margarine manufacture.

SLO: Ch4 p.2 — describe transition elements as metals that act as catalysts for industrial purposes (example: Haber process). Source: Ch4 p.64, §4.5, point 5(a) — "Iron is used in the Haber Process for the synthesis of ammonia."

Q15. Transition elements have high densities mainly because of their: → **C – higher atomic masses and closely packed structures** **1 mark**

Transition elements have **higher atomic masses** combined with **closely packed metallic structures**, giving high densities, e.g. iron = 7.87 g/cm³, tungsten = 19.3 g/cm³.

SLO: Ch4 p.2 — describe transition elements as metals that have high densities. Source: Ch4 p.64, §4.5, point 1 "High Density".

Q16. A covalent bond in which BOTH shared electrons come from only ONE atom is a: → **D – Coordinate covalent bond** **1 mark**

In a normal covalent bond, each atom contributes one electron to the shared pair. In a **coordinate (dative) covalent bond**, one atom donates BOTH electrons of the shared pair, while the other atom donates none.

SLO: Ch5 p.1 — define ionic, covalent, coordinate covalent and metallic bonds.

Q17. When potassium reacts with chlorine to form KCl, the potassium atom: → **A – loses one electron to become K⁺**

1 mark

Potassium (2,8,8,1) has one valence electron. To attain a stable noble-gas configuration it **loses this one electron**, forming K⁺, which is then attracted to the Cl⁻ ion formed when chlorine gains that electron.

SLO: Ch5 p.1 — compare between the formation of cations and anions.

Q18. Metals conduct electricity well mainly because they contain: → **B – a 'sea' of delocalized electrons that can move freely through the structure** **1 mark**

In metallic bonding, the outer electrons are not tied to any one atom but form a **'sea' of delocalized electrons** that can move freely, allowing metals to conduct electricity.

SLO: Ch5 p.2 — justify the availability of free charged particles (electrons or ions) for conduction of electricity in metallic bonds.

Q19. Diamond has an extremely high melting point mainly because it: → **C – has a giant covalent (network) structure of countless strong covalent bonds** **1 mark**

Diamond is not made of separate small molecules; every carbon atom is joined to four others by strong covalent bonds throughout the crystal, forming a **giant covalent (network) structure** that requires enormous energy to break apart.

SLO: Ch5 p.2 — interpret the strength of forces of attraction and their impact on melting/boiling points; justify the suitability of usage of diamond (e.g. in cutting tools).

Q20. One mole of any substance contains Avogadro's number of particles, which equals: → **D – 6.022×10^{23}** **1 mark**

Avogadro's number, 6.022×10^{23} , is the number of particles (atoms, molecules or formula units) present in exactly one mole of any substance.

SLO: Ch6 p.1 — define mole as the amount of substance containing Avogadro's number (6.022×10^{23}) of particles.

Q21. Moles present in 2.3 g of sodium metal (Na = 23): → **B – 0.1 mol** **1 mark**

moles = mass ÷ molar mass = $2.3 \div 23 = 0.1 \text{ mol}$.

SLO: Ch6 p.1 — use the relationship amount of substance = mass / molar mass to calculate number of moles.

Q22. Which has an empirical formula DIFFERENT from its molecular formula? → **A – Benzene, C₆H₆ (empirical CH)**

1 mark

Benzene's molecular formula C₆H₆ simplifies to the empirical formula **CH** (ratio 1:1), so the two differ. Water, ammonia and carbon monoxide already show the simplest whole-number ratio of atoms, so their empirical and molecular formulas are identical.

SLO: Ch6 p.1 — define molecular formula and empirical formula of a compound.

Q23. Coefficients for $\text{H}_2\text{O}_2 \rightarrow \text{H}_2\text{O} + \text{O}_2$: → **D – 2, 2, 1** **1 mark**

$2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$. Check: left has 4 H and 4 O; right has 4 H (in 2H₂O) and 2+2 = 4 O (2 from H₂O, 2 from O₂). **Balanced.**

SLO: Ch6 p.1 — construct chemical equations to show reactants forming products, including balancing.

Q24. Oxidation number of sulfur in H₂SO₄: → **C – +6** **1 mark**

$2(+1) + x + 4(-2) = 0 \rightarrow 2 + x - 8 = 0 \rightarrow x = +6$.

SLO: Ch7 p.1 — use roman numerals / rules to indicate the oxidation number of an element in a compound.

Q25. In a redox reaction, the oxidizing agent is the substance that: → **B – gains electrons and is itself reduced** **1 mark**

An **oxidizing agent** takes electrons from another substance (causing that substance to be oxidized) and, in doing so, gains electrons itself and is reduced.

SLO: Ch7 p.1 — identify oxidizing and reducing agents in redox reactions.

Q26. Oxidation number of chromium in the chromate ion, CrO₄²⁻: → **A – +6** **1 mark**

Sum of oxidation numbers = charge on ion: $x + 4(-2) = -2 \rightarrow x - 8 = -2 \rightarrow x = +6$.

SLO: Ch7 p.1 — explain that the sum of the oxidation numbers in an ion is equal to the charge on the ion.

Q27. A catalyst speeds up a chemical reaction mainly by: → **C – providing an alternative reaction pathway with lower activation energy** **1 mark**

A catalyst offers a different reaction route (mechanism) with a **lower activation energy**, so more colliding particles have enough energy to react successfully, speeding up the reaction without being used up.

SLO: Ch8 p.1 — explain that activation energy depends on reaction pathway, which can be changed using a catalyst.

Q28. In an exothermic reaction, the enthalpy change (ΔH) is: → **D – negative, because energy is released to the surroundings** **1 mark**

By convention, **exothermic** reactions release energy to the surroundings, so ΔH is given a **negative** sign; endothermic reactions absorb energy and have a positive ΔH .

SLO: Ch8 p.1 — state that thermal energy is called enthalpy change and recognize its sign as negative for exothermic and positive for endothermic reactions.

Q29. A reaction that proceeds both forward and reverse under the same conditions and never completes is a: → **A – Reversible reaction** **1 mark**

A **reversible reaction** is shown with a double arrow ($\rightleftharpoons$) and proceeds in both directions simultaneously, so the reactants are never fully converted to products.

SLO: Ch9 p.1 — recognize that reversible reactions are shown by $\rightleftharpoons$ and may not go to completion.

Q30. Colour change when hydrated $\text{CoCl}_2 \cdot 6\text{H}_2\text{O}$ is heated and loses water of crystallization: → **B – pink to blue** **1 mark**

Hydrated cobalt(II) chloride is **pink**; on heating it loses its water of crystallization and turns into **blue** anhydrous cobalt(II) chloride. Adding water reverses this change (blue back to pink).

SLO: Ch9 p.1 — describe the effect of heat on hydrated compounds, particularly cobalt(II) chloride. Source: Ch9 Key Points, p.138 — "Anhydrous cobalt(II) chloride is a blue solid... Hydrated cobalt(II) chloride is a pink solid."

SECTION B — Short Questions ($2 \times 4 = 8$ Marks)

2(i) Write a balanced equation, with state symbols, for the burning of hydrogen in oxygen to produce water. State whether it is exothermic or endothermic, with a reason. **4 marks**

Word equation: hydrogen + oxygen → water **—**

Balanced symbol equation with state symbols:

$2\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{H}_2\text{O}(\text{l})$ **1 mark** (correct state symbols) **1 mark**

Check: left has 4 H, 2 O; right has 4 H (in $2\text{H}_2\text{O}$), 2 O. Balanced.

Classification: This is an **exothermic** reaction (energy is released to the surroundings, ΔH is negative). **1 mark**

Reason: Breaking the H–H and O=O bonds in the reactants absorbs energy, but forming the new O–H bonds in water releases MORE energy than was absorbed. Since bond making releases more energy than bond breaking absorbs, there is a net release of energy to the surroundings. **1 mark**

SLO: Ch6 p.1 (construct and balance chemical equations, including state symbols) and Ch8 p.1 (differentiate exothermic/endothermic; recognize that bond breaking is endothermic and bond making is exothermic).

2(ii) Derive the formula of aluminium sulfate from Al^{3+} and SO_4^{2-} , using the criss-cross method. **4 marks**

Step 1 – identify the charges (valencies): Aluminium ion = Al^{3+} (charge +3); Sulfate ion = SO_4^{2-} (charge -2).

1 mark

Step 2 – criss-cross / LCM method: The charges must balance to zero overall. LCM of 3 and 2 = 6. To get +6 we need 2 aluminium ions ($2 \times 3 = 6$); to get -6 we need 3 sulfate ions ($3 \times 2 = 6$). So the ratio of $\text{Al}^{3+} : \text{SO}_4^{2-}$ is 2 : 3. **1 mark**

Step 3 – write the formula: $\text{Al}_2(\text{SO}_4)_3$ **1 mark**

Step 4 – brackets: Since sulfate (SO_4^{2-}) is a polyatomic ion and more than one of it is needed, it must be enclosed in brackets before adding the subscript 3, giving the correct notation $\text{Al}_2(\text{SO}_4)_3$ rather than the incorrect "AlSO4323".

1 mark

SLO: Ch7 p.1 — derive the formula of ionic compounds from ionic charges and oxidation numbers.

SECTION C — Long Question ($1 \times 8 = 8$ Marks)

3. Compound Z: 49.3% C, 6.8% H, 43.9% O by mass; relative molecular mass = 146. 8 marks

(a) Empirical formula 3 marks

Moles of C = $49.3 \div 12 = 4.108$

Moles of H = $6.8 \div 1 = 6.8$

Moles of O = $43.9 \div 16 = 2.744$ 1 mark

Divide each by the smallest (2.744): C = $4.108/2.744 = 1.5$; H = $6.8/2.744 = 2.5$; O = $2.744/2.744 = 1$

Ratio C : H : O = 1.5 : 2.5 : 1. Multiply throughout by 2 to clear the fractions: **C : H : O = 3 : 5 : 2** 1 mark

Empirical formula = **C₃H₅O₂** 1 mark

(b) Molecular formula 1 mark

Empirical formula mass = $3(12) + 5(1) + 2(16) = 36 + 5 + 32 = 73$

n = relative molecular mass $\div$ empirical formula mass = $146 \div 73 = 2$

Molecular formula = $(\text{C}_3\text{H}_5\text{O}_2) \times 2 = \text{C}_6\text{H}_{10}\text{O}_4$

(c) Moles of Z in 43.8 g 1 mark

moles = mass $\div$ molar mass = $43.8 \div 146 = 0.3 \text{ mol}$

(d) Number of molecules of Z 1 mark

molecules = moles $\times N_0 = 0.3 \times 6.022 \times 10^{23} = 1.807 \times 10^{23}$ molecules (accept 1.8×10^{23})

(e) Balanced combustion equation 1 mark

$2 \text{C}_6\text{H}_{10}\text{O}_4(\text{l}) + 13 \text{O}_2(\text{g}) \rightarrow 12 \text{CO}_2(\text{g}) + 10 \text{H}_2\text{O}(\text{l})$

Check: left = 12 C, 20 H, $4 \times 2 + 13 \times 2 = 8 + 26 = 34$ O; right = 12 C, 20 H, $12 \times 2 + 10 \times 1 = 24 + 10 = 34$ O. Balanced.

(f) Mass of CO₂ from 0.3 mol of Z 1 mark

Mole ratio Z : CO₂ = 2 : 12 = 1 : 6. So 0.3 mol Z produces $0.3 \times 6 = 1.8 \text{ mol CO}_2$.

Mass of CO₂ = moles $\times$ molar mass = $1.8 \times 44 = 79.2 \text{ g}$

SLO: Ch6 p.1 — define empirical/molecular formula; use amount of substance = mass/molar mass; define mole and Avogadro's constant; construct and balance chemical equations.

— End of Answer Key —